

VIVEK KUMAR MISHRA

O1-A Petition for Alien of Extraordinary Ability

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[LinkedIn Profile](#) | [Portfolio Website](#)

PROFILE SUMMARY

Associate Software Engineer and IEEE Senior Member with deep expertise in backend systems, distributed architecture, and AI/ML engineering. Specialized in designing scalable microservices and RESTful APIs using Java, Spring Boot, and AWS cloud infrastructure. Proven track record building production systems serving 80+ million users at JPMorgan Chase with 99.9% reliability. Strong foundation in machine learning methodologies, with research permanently archived in U.S. National Transportation Library. Active contributor to AI/ML research community as peer reviewer for NeurIPS 2025 and ICLR 2025.

EDUCATION

Master of Science in Computer Science

August 2021

California State University, Long Beach (CSULB) – Long Beach, CA | GPA: 3.8/4.0

Bachelor of Science in Information Technology

May 2014

West Bengal University of Technology (now MAKAUT) – Kolkata, India

PROFESSIONAL EXPERIENCE

Associate Software Engineer

October 2021 – Present

JPMorgan Chase & Co. – Wilmington, Delaware

- Founding engineer on Chase Shopping Hub platform, architecting backend APIs and microservices for 80+ million Chase cardmembers
- Designed scalable RESTful APIs using Java, Spring Boot, and AWS (DynamoDB, SQS, ECS, S3) processing 5+ million customer requests monthly with 99.9% reliability
- Improved application performance by 43% through backend microservices optimization
- Increased automated test coverage from 17% to 97% using JUnit and Mockito

Research Assistant

March 2020 – August 2021

California State University, Long Beach – Long Beach, CA

- Recruited for specialized Python and machine learning expertise for federally-funded intelligent transportation systems research (U.S. Department of Transportation via Mineta Transportation Institute)
- Developed machine learning methodologies for transportation data analysis using NumPy, Pandas, TensorFlow, and NLP
- Authored research papers permanently archived in U.S. National Transportation Library (ROSA P Records dot/61210, dot/59030)

- Research methodologies subsequently adopted by Florida Department of Transportation in federally-funded infrastructure programs

Member of Technical Staff

July 2018 – January 2020

Oracle Financial Services Software Limited – Bangalore, India

- Automated Oracle Cloud Infrastructure (OCI) testing workflows using Ansible, Python, and Go
- Developed infrastructure-as-code solutions using Terraform and Docker for distributed cloud systems
- Commanded salary 83% above market average in 2018 (Rs.11,00,000 vs. Rs.6,00,000) based on exceptional technical capabilities

Programmer Analyst

August 2014 – July 2018

Cognizant Technology Solutions – Bangalore, India

- Automated 500+ test cases using Python and shell scripting, reducing CI runtime from 4.5 to 2.5 hours
- Earned Global Project of the Year Award (2x recipient) for outstanding technical contributions

O1-A CRITERIA SATISFIED (6 out of 8)

1. Original Contribution of Major Significance

- **Technical Book:** “Design Patterns for AI Agents” (ISBN: 979-8268912371) – adopted by AICTE-approved institutions and integrated into government engineering curriculum
- **JPMorgan Chase:** Founding engineer included in architectural design of Chase Shopping Hub, designing backend systems processing 5M+ requests monthly
- **Federal Impact:** ML methodologies adopted by Florida DOT and permanently indexed in U.S. National Transportation Library (ROSA P Records dot/61210, dot/59030)

2. Scholarly Articles [[Evidence](#)]

8 peer-reviewed publications – 37 total citations, h-index: 3 (Google Scholar)

- “Integrating Solar Photovoltaic Systems into the Grid: An Overview of AI Application” – *IEEE Access* (Q1, IF 3.9)
- “Framework for Rail Transport Inequality Assessment” – *MDPI Sustainability* (Q2, IF 3.9) – **15 citations**
- “Evaluating Financing Mechanisms and Economic Benefits to Fund Grade Separation Projects” – **12 citations**
- 2 publications permanently archived in U.S. National Transportation Library

3. Media Recognition [[Evidence](#)]

- **IEEE Official Recognition:** Featured as “IEEE Young Creator” on IEEE Transmitter, IEEE.tv, and IEEE’s official Twitter/X (342,900 followers)
- **Major Press:** Dainik Bhaskar (63M readers), Rajasthan Patrika (25.9M readers), First India News (4.4M+ YouTube subscribers)
- **Expert Quotations:** TechTarget (1.4M+ daily readers), Dice.com (2.4M monthly visitors), TechBullion (230K monthly), VKTR.com (7M+ readers)

4. Judging Work of Others [[Evidence](#)]

- **NeurIPS 2025** Position Paper Track Reviewer (24.52% acceptance rate, H5-index 278)
- **ICLR 2025** FM-Wild Workshop Reviewer (32% acceptance rate, H5-index 362)
- **IEEE Senior Member Application Review Panel** – June 2025
- Journal reviewer: Informatica, Virtus InterPress (Risk Governance, Corporate & Business Strategy)
- Hackathon judge: PennApps (University of Pennsylvania), DragonHacks 2024, Claro 2024

5. Membership in Distinguished Associations [[Evidence](#)]

- **IEEE Senior Member** (Member #100495418) – November 2024
- Requires 10 years professional practice and 5 years of “significant performance” evaluated by three IEEE Senior Members/Fellows
- Only 8% of IEEE’s 460,000+ members achieve this distinction

6. Critical Role for Distinguished Organizations

- **JPMorgan Chase & Co.** (America’s largest bank, \$4.1+ trillion assets, 84M customers)
- Founding engineer with architectural decision-making authority on Chase Shopping Hub platform
- Designed mission-critical infrastructure serving 80+ million Chase cardmembers

HONORS & AWARDS

- Global Recognition Award 2024
- #FutureOfMeterReading Contest Winner – Capgemini Super Techies
- Global Project of the Year Award (2x) – Cognizant Technology Solutions

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, Go, Shell Scripting

Frameworks: Spring Boot, Spring Security, React, TensorFlow, NumPy, Pandas

Cloud & Infrastructure: AWS (VPC, DynamoDB, IAM, SQS, ECS, S3), Terraform, Docker, Kubernetes, Ansible

AI/ML Tools: LLMs, GenAI, OpenCV, NLP

Security: OAuth 2.0, OIDC, SAML, JWT